

Choosing and Combining Factoring Techniques

Common factor first, monic and non-monic trinomials, and differences of squares

- The expressions below are **mixed on purpose**. Nobody tells you which method to use — deciding that is the skill this sheet practises.
 - Run the same three questions on every problem, in this order: **(1)** Is there a common factor in every term? **(2)** How many terms are left? **(3)** Which pattern fits what is left?
 - **Factor completely.** You are not finished while something inside the parentheses still factors.
 - Check any answer by multiplying it back out.
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1. Factor completely: $6x^2 + 9x$

Answer: _____

2. Factor completely: $x^2 + 7x + 12$

Answer: _____

3. Factor completely: $x^2 - 64$

Answer: _____

4. Factor completely: $x^2 - 3x - 10$

Answer: _____

5. Factor completely: $2x^2 + 7x + 3$

Answer: _____

6. Factor completely: $4x^2 - 36$

Answer: _____

7. Factor completely: $49x^2 - 25$

Answer: _____

8. Factor completely: $3x^2 - 10x + 8$

Answer: _____

9. Factor completely: $x^2 - 11x + 24$

Answer: _____

10. A classmate says that $2x^2 + 3x - 20$ factors as $(2x - 5)(x + 4)$. Multiply the two binomials out and write the product in standard form, so that the claim can be compared with the original expression.

Answer: _____

11. Factor completely: $2x^3 - 8x^2 - 24x$

Answer: _____

12. Synthesis challenge. $x^4 - 13x^2 + 36$

- (a) Factor the expression completely. Write the fully factored form on the answer line.
- (b) In the space above, explain which technique you used *first* and why that technique was available, then name the technique that finished the job.

Answer: _____